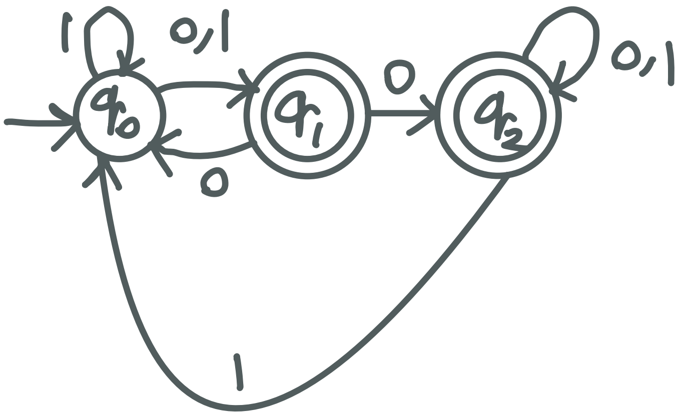
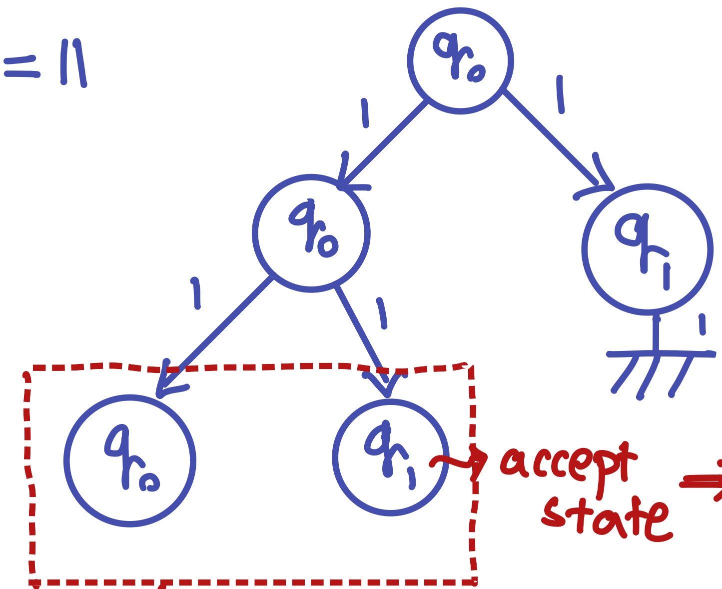


Task 1

Input $w = 11$

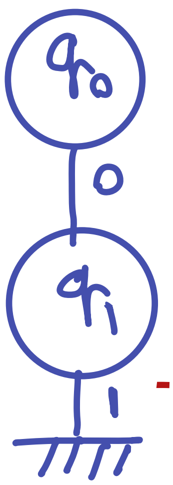


no transition for 1

accept state $\Rightarrow$ $w = 11$ is <u>accepted</u>

set of NFA states after processing $w = 11$

Input $w = 01$



no transitions for 1

This set doesn't contain an accept state $\Leftarrow$ set of NFA states after processing $w = 01$ $= \emptyset$

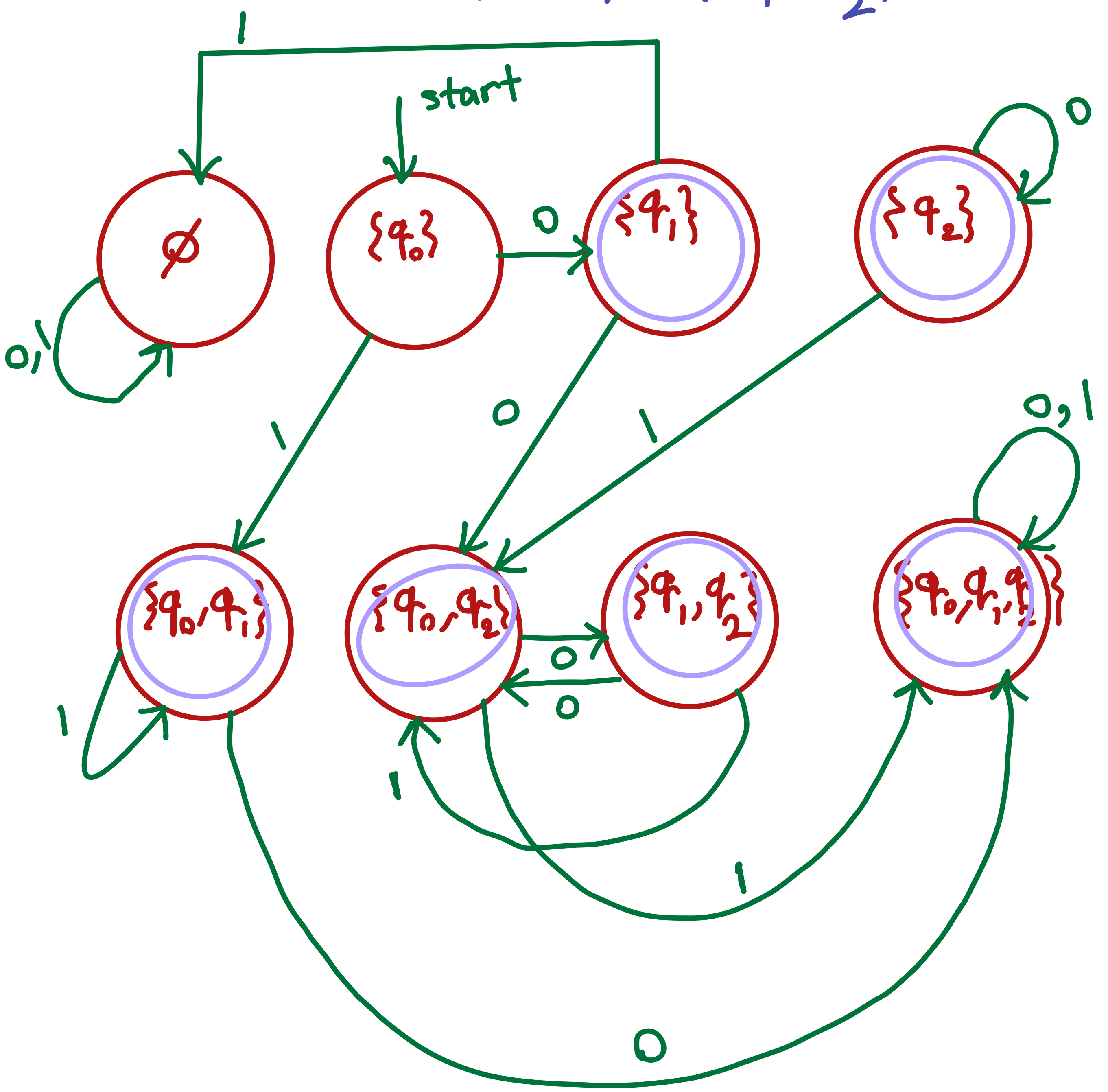
$\Rightarrow$ $w = 01$ is rejected

$Q_{NFA} = \{q_0, q_1, q_2\}$

$Q_{DFA} = P(Q_{NFA})$

↳ power set

$= \{\emptyset, \{q_0\}, \{q_1\}, \{q_2\}, \{q_0, q_1\}, \{q_0, q_2\}, \{q_1, q_2\}, \{q_0, q_1, q_2\}\}$



$w = 11$

$$\{q_0\} \xrightarrow{1} \{q_0, q_1\} \xrightarrow{1} \underbrace{\{q_0, q_1\}}_{\text{accept state}} \Rightarrow \text{accepted}$$

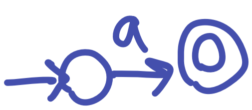
$w = 01$

$$\{q_0\} \xrightarrow{0} \{q_1\} \xrightarrow{1} \underbrace{\emptyset}_{\text{not an accept state}} \rightarrow \text{rejected}$$

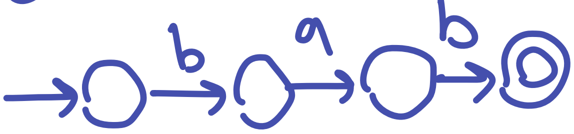
Task 2

$$ab(a \cup bab)^* \cup bbb$$

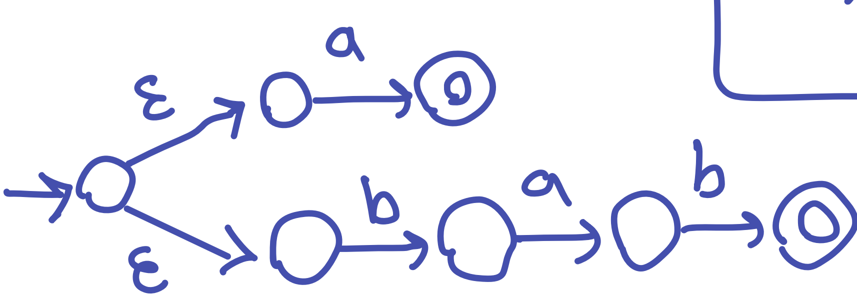
a :



bab :

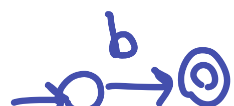


a ∪ bab :



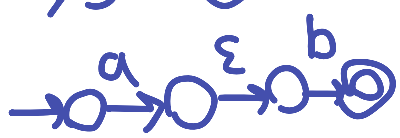
OR

b :

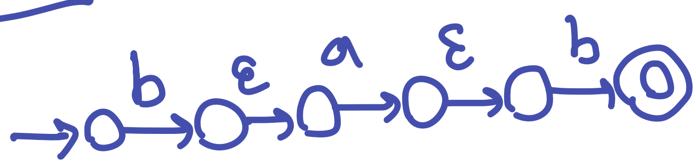


a :

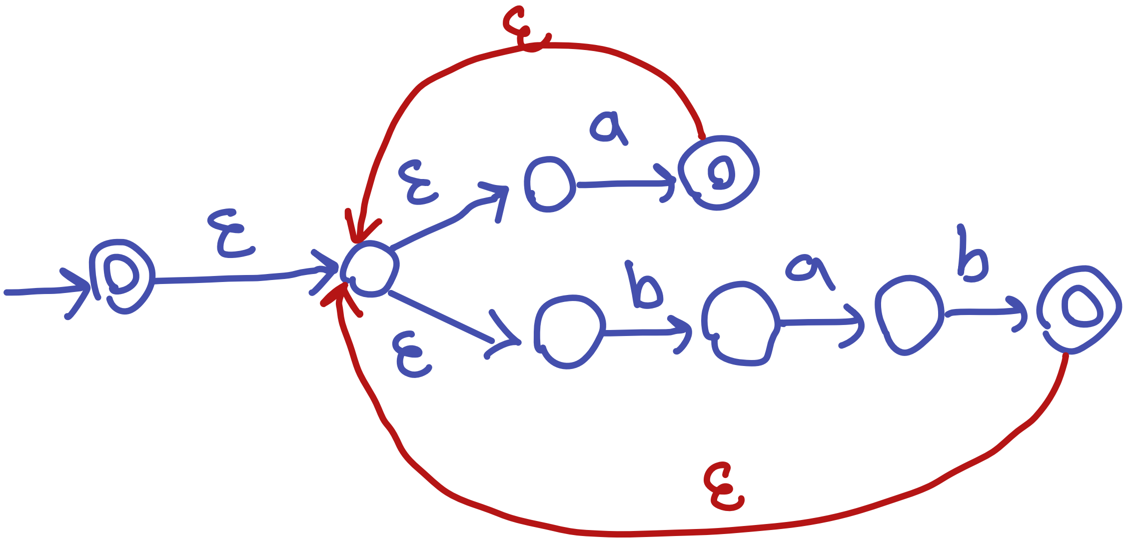
ab :



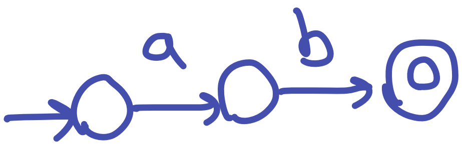
bab :



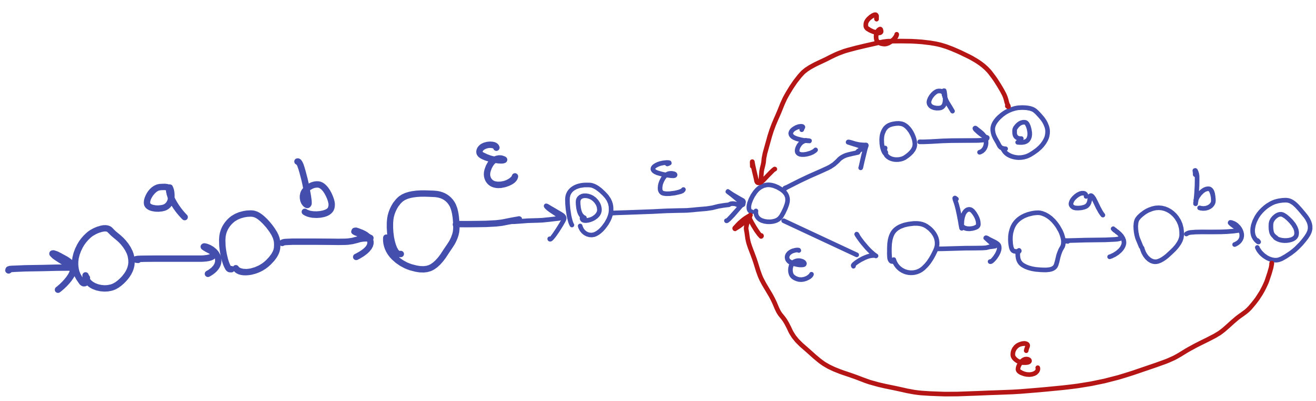
$(a \cup bab)^*$:



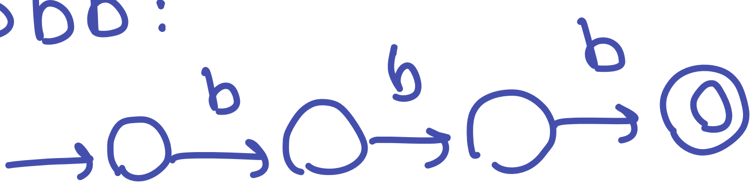
ab:



$ab(a \cup bab)^*$:

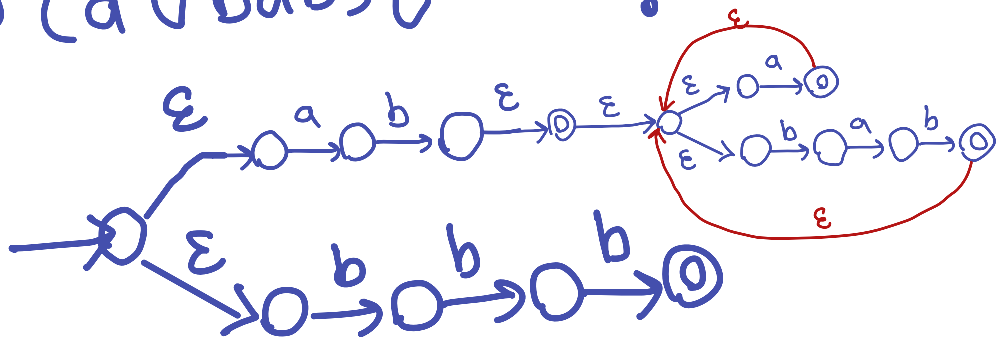


bbb:



OR, we could also make this using concat. construction

$ab(a \cup bab)^* \cup bbb$:



Task 3

$L = \{ ww \mid w \in \{0,1\}^* \}$

Suppose L is regular $\Longrightarrow$ pumping Lemma applies

pick string $s = 0^p 1^p 0^p 1^p \in L$

where $p \geq 1$ is the pumping length

$|s| = 4p \geq p \Longrightarrow$ it's a valid choice for pumping Lemma

According to Pumping Lemma :

$\exists\ s = xyz$ s.t. $|xy| \leq p$, $|y| \geq 1$, $xy^iz \in L$

$$s = \boxed{\overbrace{0\cdots0}^{p}}\;\overbrace{1\cdots1}^{p}\;\overbrace{0\cdots0}^{p}\;\overbrace{1\cdots1}^{p}$$

$|xy| \leq p \Longrightarrow$ xy can only contain zeros

$|y| \geq 1 \Longrightarrow$ y contains one ore more "0"'s.

$\Longrightarrow$ $y = 0^k$ for some $1 \leq k \leq p$

$\Longrightarrow$ $xy^iz = xy^0z = xz = 0^{p-k}1^p0^p1^p \notin L$

(for $i=0$)

because if you try to divide $0^{p-k}1^p0^p1^p$ in two parts (ww)

The first part ends with "0"

but the second

part ends with "1"

$\Rightarrow$ Contradiction

$\Rightarrow$ L is not regular